

LLM-Guided Reinforcement Learning For Maritime Ship Collision Avoidance

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I. INTRODUCTION

Safe and efficient collision avoidance is a critical requirement for autonomous maritime navigation in multi-vessel encounter scenarios governed by the International Regulations for Preventing Collisions at Sea (COLREGs) [1]. Classical geometric, optimization-based, and rule-based approaches have demonstrated effectiveness in structured settings, but can often struggle to scale in dynamic, increasingly dense, multi-agent environments [2]–[4]. Reinforcement learning (RL) methods have emerged as a promising alternative due to their scalability and robustness to environmental uncertainty [5]–[8]. However, RL methods often produce black-box policies whose learned representations resist formal verification and often lack interpretability.

Large-language models (LLMs) enable interpretable reasoning aligned with rule-based frameworks and have recently been explored for more verifiable COLREGs-compliant decision making [9]. Despite this advantage, LLM-based systems can behave unreliably. So, they require external tools to ensure safe operation. However, these tools, particularly rule-based planners, limit adaptability and scalability in complex environments.

This work presents an LLM-guided reinforcement learning (RL) architecture for maritime ship collision avoidance by integrating a learned RL-based local planner into the CORALL (COLREGs-Guided Risk-Aware LLM) framework [9]. As visualized in Fig. 1, the proposed approach replaces the rule-based reactive collision avoidance local planner module with a reinforcement learning policy conditioned on high-level navigation intent, thus preserving COLREGs-compliant high-level decision-making. The proposed architecture improves operation efficiency and robustness to uncertainty, while still maintaining safe separation behavior in multi-agent encounter scenarios.

II. METHODOLOGY

In our work, the rule-based reactive collision avoidance planner within CORALL is replaced with a reinforcement learning (RL) policy that controls vessel heading in a simulated maritime environment governed by first-order vessel dynamics. Like CORALL, our proposed architecture employs an LLM to interpret maritime encounter context and generate maneuver intent consistent with COLREGs [9]. The

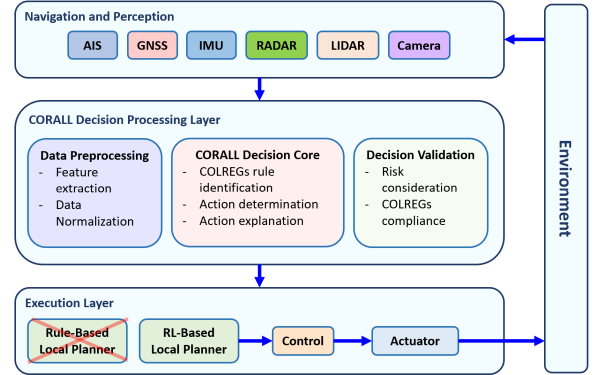


Fig. 1. Proposed update to local planner within baseline CORALL architecture.

learned policy operates at the execution layer, translating this intent into continuous control actions. In contrast to the RL-based planner, the baseline rule-based local planner computes pairwise collision risk indices using Z-shaped membership functions to generate collision-avoidance maneuvers [9].

The RL policy is trained using Proximal Policy Optimization (PPO) on simulated vessel encounters of variable scale (2-4 agents in the environment) [10]. The reward function is designed to balance trajectory tracking, collision avoidance, and risk-aware behavior as shown in Equation (1). Here terms correspond to path-following progress, risk-exposure, collision penalties, minimum separation distance enforcement, and waypoint completion.

$$r_{\text{total}} = r_{\text{along}} + r_{\text{risk}} + r_{\text{collision}} + r_{\text{separation}} + r_{\text{success}} \quad (1)$$

III. PRELIMINARY RESULTS

Performance is evaluated against the original CORALL rule-based baseline planner across predefined Imazu encounter scenarios ranging between two to four total vessels in the environment (including the ownship) [9]. Evaluation metrics include collision rate, minimum separation distance, trajectory efficiency (total path length and mission completion time), closest point of approach (CPA), and time-weighted risk exposure derived from CPA-based risk indices [11].

Results in Fig. 2 indicate that the RL-based planner reduces total path length by up to 14% and the total time traveled up to

a 45% in multi-agent scenarios. Significant reductions in time-weighted risk exposure in the two and four-agent encounters is also observed. Enhanced performance is maintained while maintaining a minimum separation guideline of approximately three vessel lengths as indicated by the minimum separation threshold shown in Fig. 2. In each scenario, all ships in the environment are initialized with an initial speed of 9.5 m s^{-1} with different initial headings.

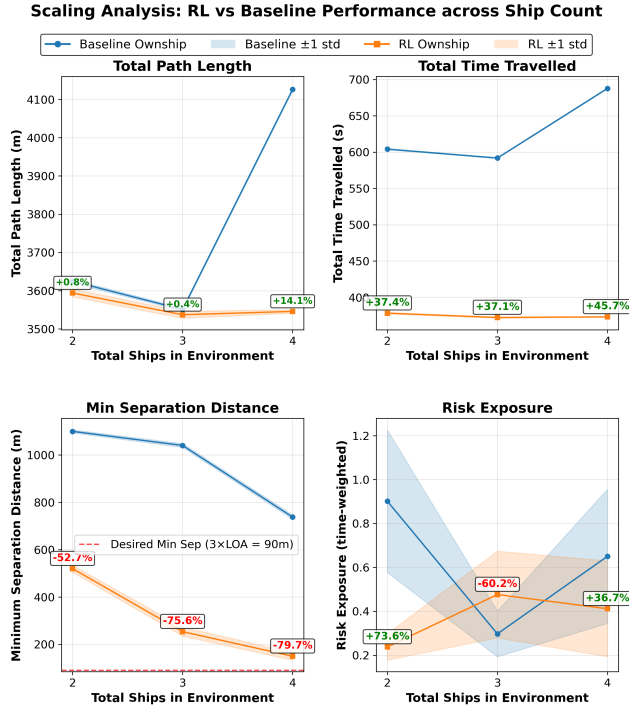


Fig. 2. Aggregate performance trends between RL-based and rule-based ownship planners across increasing number of agents 2–4 vessel scenarios evaluated over 50 simulations each

The qualitative trajectory results in Fig. 3 show how the RL-guided ownship demonstrates successful navigation on scaled representative scenarios of increasingly congested environments without collision while adapting maneuver strategies relative to the rule-based planner, despite some trajectory variances between the two approaches.

IV. SUMMARY AND EXPECTED OUTCOMES

This work demonstrates the feasibility of integrating reinforcement learning into a hierarchical, LLM-guided framework for collision avoidance. The newly proposed architecture preserved interpretable, COLREGs-compliant decision-making while improving trajectory efficiency and reducing risk exposure. The findings highlight important tradeoffs between safety, efficiency, and scalability in multi-agent environments.

In our full paper, we expect to publish results on refining reward design, improving policy robustness in denser scenarios, extending the framework toward multi-agent control for scalable deployment, and exploring better task allocation between the LLM and RL agent. This approach provides a

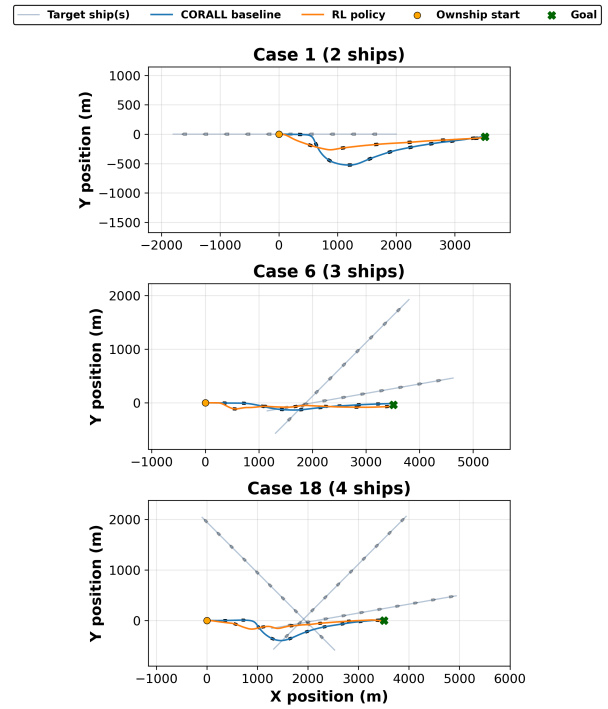


Fig. 3. Ownship trajectory comparison across representative Imazu scenarios.

pathway towards interpretable, learning-enabled autonomy in safety-critical maritime systems.

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